



II Sentinel

AI-assisted infrastructure monitoring for African industry

The failure warnings Western plants pay millions for — at \$80 a sensor, built for African n

THE PROBLEM

One hour of a stalled mine pump costs \$150,000+. Operators juggle 3-6 disconnected tools, and the oldest equipment — most likely to fail — carries zero telemetry because SCADA points cost \$5,000+ each.

THE PRODUCT

One live map over mining OT, telecom power, and network assets. AI health scoring that learns each machine's normal and flags deviation days before thresholds trip; time-to-failure estimates; SMS/WhatsApp alerts that work on 2G. Every score decomposes into readable factors — auditable, never a black box. \$80 Raspberry Pi edge nodes put sensors on 1990s equipment; offline buffering survives dropped links.

WHY NOW, WHY HERE

Built for constraints imported tools ignore: intermittent connectivity, feature-phone alerting, in-region hosting (Johannesburg, ~30ms from Harare). Beachhead: Zimbabwe mining OT. Expansion: tower power for telecoms (generator fuel theft — an existing budget line), then SADC. Moat: an anonymized cross-site failure dataset for African industry — the software is copyable; the data never will be.

BUSINESS MODEL

Mining OT Edition \$8-15/asset/month (per-shaft bundles) · Pi nodes ~\$120 installed · paid deployment. 30-day pilot: 10 assets, ~\$500 or free against a signed LOI, success metrics agreed in writing (alert precision $\geq 70\%$, ≥ 1 failure flagged before the crew, < 2 false positives per device-week).

STATUS

Working platform deployed today: live digital-twin map, explainable AI, cascade failure simulation, real ICMP LAN monitoring, modular editions, Pi collector kit, safety-advisory CBS integrity monitor.

THE ASK One 30-day pilot site and one letter of intent.

Ten assets, one shaft. If we don't catch a failure before your crew does, you owe nothing.